

Docker - Docker Compose

Create an initial docker_compose.yml file:

```
docker-compose.yml M
datascientest-data-engineering > docker > docker_compose_exercise > docker-compose.yml >
1  version: '3.9'
   > Run All Services
2  services:
   > Run Service
3  jupyter:
4  image: jupyter/minimal-notebook:ubuntu-18.04
   > Run Service
5  elasticsearch:
6  image: elasticsearch:7.2.0
7
```

```
ubuntu@ip-172-31-18-113:~/datascientest-data-engineering/docker/docker_compose_exercise$ ls
docker-compose.yml
ubuntu@ip-172-31-18-113:~/datascientest-data-engineering/docker/docker_compose_exercise$ docker-compose up
Building with native build. Learn about native build in Compose here: https://docs.docker.com/go/compose-native-build/
Creating network "docker_compose_exercise_default" with the default driver
Pulling jupyter (jupyter/minimal-notebook:ubuntu-18.04)...
ubuntu-18.04: Pulling from jupyter/minimal-notebook
23884877105a: Pull complete
bc38caa0f5b9: Pull complete
2910811b6c42: Pull complete
36505266dccc: Pull complete
7c7b63c8c081: Pull complete
4590b6ea70f4: Pull complete
6164b0017070: Pull complete
9e37625b57c1: Pull complete
605292fc55ae: Pull complete
998e0039cb74: Pull complete
dbaceb237cab: Pull complete
d1a9cd1c299e: Pull complete
5100ff5e8ebb: Pull complete
94bb066d5eea: Pull complete
dd043717b506: Pull complete
5b60bba7ac08: Pull complete
61a642f52604: Pull complete
Digest: sha256:7cb9ef93b906c90de8257872555a6a4c1b50999cac0a72286e459f9dc91af570
Status: Downloaded newer image for jupyter/minimal-notebook:ubuntu-18.04
Pulling elasticsearch (elasticsearch:7.2.0)...
7.2.0: Pulling from library/elasticsearch
8ba884070f61: Pull complete
2211b14f8b24: Pull complete
617ccdb47f3d: Pull complete
915ae6b2c338: Pull complete
b414b7f29a7d: Pull complete
547bddd35d62: Pull complete
8353a2ed248c: Pull complete
Digest: sha256:84b5bc2fd15b0f1f5bf78c8c6ee34b6ae5a46ab81be1c2cfa678eea0c6457a46
Status: Downloaded newer image for elasticsearch:7.2.0
Creating docker_compose_exercise_elasticsearch_1 ... done
Creating docker_compose_exercise_jupyter_1 ... done
Attaching to docker_compose_exercise_jupyter_1, docker_compose_exercise_elasticsearch_1
jupyter_1 | Executing the command: jupyter notebook
jupyter_1 | [I 08:52:49.025 NotebookApp] Writing notebook server cookie secret to /home/jovyan/.local/share/jupy
ter/runtime/notebook_cookie_secret
elasticsearch_1 | OpenJDK 64-Bit Server VM warning: Option UseConcMarkSweepGC was deprecated in version 9.0 and will l
ikely be removed in a future release.
jupyter_1 | [I 08:52:51.600 NotebookApp] JupyterLab extension loaded from /opt/conda/lib/python3.7/site-packages
/jupyterlab
jupyter_1 | [I 08:52:51.601 NotebookApp] JupyterLab application directory is /opt/conda/share/jupyter/lab
jupyter_1 | [I 08:52:51.606 NotebookApp] Serving notebooks from local directory: /home/jovyan
jupyter_1 | [I 08:52:51.606 NotebookApp] The Jupyter Notebook is running at:
jupyter_1 | [I 08:52:51.608 NotebookApp] http://06d580044899:8888/?token=564693e586a711330dc2ec179db112d174f481d
45761ed90
jupyter_1 | [I 08:52:51.608 NotebookApp] or http://127.0.0.1:8888/?token=564693e586a711330dc2ec179db112d174f481
d45761ed90
jupyter_1 | [I 08:52:51.609 NotebookApp] Use Control-C to stop this server and shut down all kernels (twice to s
kip confirmation).
jupyter_1 | [C 08:52:51.620 NotebookApp]
jupyter_1 | To access the notebook, open this file in a browser:
jupyter_1 | file:///home/jovyan/.local/share/jupyter/runtime/nbserver-7-open.html
jupyter_1 | Or copy and paste one of these URLs:
jupyter_1 | http://06d580044899:8888/?token=564693e586a711330dc2ec179db112d174f481d45761ed90
jupyter_1 | or http://127.0.0.1:8888/?token=564693e586a711330dc2ec179db112d174f481d45761ed90
```

Check status and stop Docker container in new terminal:

```
ubuntu@ip-172-31-18-113:~/datascientest-data-engineering/docker/docker_compose_exercise$ docker ps
CONTAINER ID   IMAGE                                COMMAND                  NAMES                  CREATED        STATUS
PORTS
06d580044899   jupyter/minimal-notebook:ubuntu-18.04 "tini -g -- start-no..." 3 minutes ago   Up 3 minutes
8888/tcp
d6cc0b428d28   postgres:16-alpine                "docker-entrypoint.s..." 6 weeks ago    Up 15 minutes
0.0.0.0:5432->5432/tcp, :::5432->5432/tcp
pg_container
475951b12750   dpape/pgadmin4                    "/entrypoint.sh"         6 weeks ago    Up 15 minutes
443/tcp, 0.0.0.0:5050->80/tcp, :::5050->80/tcp
pgadmin4 container
ubuntu@ip-172-31-18-113:~/datascientest-data-engineering/docker/docker_compose_exercise$

ubuntu@ip-172-31-18-113:~/datascientest-data-engineering/docker/docker_compose_exercise$ docker-compose down
Stopping docker_compose_exercise_jupyter_1 ... done
Removing docker_compose_exercise_jupyter_1 ... done
Removing docker_compose_exercise_elasticsearch_1 ... done
Removing network docker_compose_exercise_default
ubuntu@ip-172-31-18-113:~/datascientest-data-engineering/docker/docker_compose_exercise$
```

Edit docker-compose.yml file (add environment variable to ElasticSearch):

```
docker-compose.yml M X
datascientest-data-engineering > docker > docker_compose_exercise > docker-compose.yml >
2   services:
3     > Run Service
4     jupyter:
5       image: jupyter/minimal-notebook:ubuntu-18.04
6     > Run Service
7     elasticsearch:
8       image: elasticsearch:7.2.0
9       environment:
10        discovery.type: single-node
```

```
ubuntu@ip-172-31-18-113:~/datascientest-data-engineering/docker/docker_compose_exercise$ docker-compose up
Building with native build. Learn about native build in Compose here: https://docs.docker.com/go/compose-native-build/
Creating network "docker_compose_exercise_default" with the default driver
Creating docker_compose_exercise_elasticsearch_1 ... done
Creating docker_compose_exercise_jupyter_1 ... done
Attaching to docker_compose_exercise_jupyter_1, docker_compose_exercise_elasticsearch_1
jupyter_1      | Executing the command: jupyter notebook
jupyter_1      | [I 09:03:53.215 NotebookApp] Writing notebook server cookie secret to /home/jovyan/.local/share/jupyter
jupyter_1      | /runtime/notebook_cookie_secret
elasticsearch_1 | OpenJDK 64-Bit Server VM warning: Option UseConcMarkSweepGC was deprecated in version 9.0 and will like
jupyter_1      | ly be removed in a future release.
jupyter_1      | [I 09:03:54.776 NotebookApp] JupyterLab extension loaded from /opt/conda/lib/python3.7/site-packages/ju
pyterlab       |
jupyter_1      | [I 09:03:54.776 NotebookApp] JupyterLab application directory is /opt/conda/share/jupyter/lab
jupyter_1      | [I 09:03:54.782 NotebookApp] Serving notebooks from local directory: /home/jovyan
jupyter_1      | [I 09:03:54.782 NotebookApp] The Jupyter Notebook is running at:
jupyter_1      | [I 09:03:54.782 NotebookApp] http://be05704b8092:8888/?token=39abb26689cdb9e8f4c11ad595ca161dba8d4fcd36
jupyter_1      | 02c3f4
jupyter_1      | [I 09:03:54.783 NotebookApp] or http://127.0.0.1:8888/?token=39abb26689cdb9e8f4c11ad595ca161dba8d4fcd3
jupyter_1      | 602c3f4
jupyter_1      | [I 09:03:54.784 NotebookApp] Use Control-C to stop this server and shut down all kernels (twice to skip
jupyter_1      | confirmation).
jupyter_1      | [C 09:03:54.796 NotebookApp]
jupyter_1      |
jupyter_1      | To access the notebook, open this file in a browser:
jupyter_1      | file:///home/jovyan/.local/share/jupyter/runtime/nbserver-7-open.html
jupyter_1      | Or copy and paste one of these URLs:
jupyter_1      | http://be05704b8092:8888/?token=39abb26689cdb9e8f4c11ad595ca161dba8d4fcd3602c3f4
jupyter_1      | or http://127.0.0.1:8888/?token=39abb26689cdb9e8f4c11ad595ca161dba8d4fcd3602c3f4
elasticsearch_1 | {"type": "server", "timestamp": "2025-06-16T09:03:58.266+0000", "level": "INFO", "component": "o.e.No
deEnvironment", "cluster.name": "docker-cluster", "node.name": "24ecf9b0e8d2", "message": "using [1] data paths, mounts [
1/ (overlay)], net usable_space [1.8gb], net total_space [15.4gb], types [overlay]" }
```

```
ubuntu@ip-172-31-18-113:~/datascientest-data-engineering/docker/docker_compose_exercise$ docker ps
CONTAINER ID   IMAGE                                NAMES                                COMMAND                                CREATED        STATUS        PORTS
be05704b8092   jupyter/minimal-notebook:ubuntu-18.04 "tini -g -- start-no-"              2 minutes ago   Up 2 minutes   8888/tcp
24ecf9b0e8d2   elasticsearch:7.2.0                docker_compose_exercise_jupyter_1   "/usr/local/bin/dock-"              2 minutes ago   Up 2 minutes   9200/tcp,
dccc0b428d28   postgres:16-alpine                docker_compose_exercise_elasticsearch_1 "docker-entrypoint.s-"              6 weeks ago    Up 25 minutes   0.0.0.0:54
32->5432/tcp, :::5432->5432/tcp
475951b12750   dpape/pgadmin4                     pg_container                          "/entrypoint.sh"                    6 weeks ago    Up 25 minutes   443/tcp, 0
.0.0.0:5050->80/tcp, :::5050->80/tcp   pgadmin4_container
ubuntu@ip-172-31-18-113:~/datascientest-data-engineering/docker/docker_compose_exercise$ docker-compose down
Stopping docker_compose_exercise_jupyter_1 ... done
Stopping docker_compose_exercise_elasticsearch_1 ... done
Removing docker_compose_exercise_jupyter_1 ... done
Removing docker_compose_exercise_elasticsearch_1 ... done
Removing network docker_compose_exercise_default
ubuntu@ip-172-31-18-113:~/datascientest-data-engineering/docker/docker_compose_exercise$
```

Edit docker-compose.yml file again (expose Jupyter port):

```
docker-compose.yml M X
datascientest-data-engineering > docker > docker_compose_exercise >
1   version: '3.9'
2   > Run All Services
3   services:
4     > Run Service
5     jupyter:
6       image: jupyter/minimal-notebook:ubuntu-18.04
7       ports:
8         - target: 8888
9           published: 4444
10          protocol: tcp
11          mode: host
12     > Run Service
13     elasticsearch:
14       image: elasticsearch:7.2.0
15       environment:
16        discovery.type: single-node
```

```

ubuntu@ip-172-31-18-113:~/datascientest-data-engineering/docker/docker_compose_exercise$ docker-compose up
Building with native build. Learn about native build in Compose here: https://docs.docker.com/go/compose-native-build/
Creating network "docker_compose_exercise_default" with the default driver
Creating docker_compose_exercise_jupyter_1 ... done
Creating docker_compose_exercise_elasticsearch_1 ... done
Attaching to docker_compose_exercise_jupyter_1, docker_compose_exercise_elasticsearch_1
jupyter_1      | Executing the command: jupyter notebook
jupyter_1      | [I 09:12:11.900 NotebookApp] Writing notebook server cookie secret to /home/jovyan/.local/share/jupyter
/runtime/notebook_cookie_secret
jupyter_1      | [I 09:12:13.308 NotebookApp] JupyterLab extension loaded from /opt/conda/lib/python3.7/site-packages/ju
pyterlab
jupyter_1      | [I 09:12:13.310 NotebookApp] JupyterLab application directory is /opt/conda/share/jupyter/lab
jupyter_1      | [I 09:12:13.316 NotebookApp] Serving notebooks from local directory: /home/jovyan
jupyter_1      | [I 09:12:13.317 NotebookApp] The Jupyter Notebook is running at:
jupyter_1      | http://2013595c8b73:8888/?token=ac7f4a79a9dea63b36e08e179a9d7e61c084c8395f0d0f46
jupyter_1      | [I 09:12:13.318 NotebookApp] or http://127.0.0.1:8888/?token=ac7f4a79a9dea63b36e08e179a9d7e61c084c8395
f0d0f46
jupyter_1      | [I 09:12:13.319 NotebookApp] Use Control-C to stop this server and shut down all kernels (twice to skip
confirmation).
jupyter_1      | [C 09:12:13.327 NotebookApp]
jupyter_1      |
jupyter_1      | To access the notebook, open this file in a browser:
jupyter_1      | file:///home/jovyan/.local/share/jupyter/runtime/nbserver-7--open.html
jupyter_1      | Or copy and paste one of these URLs:
jupyter_1      | http://2013595c8b73:8888/?token=ac7f4a79a9dea63b36e08e179a9d7e61c084c8395f0d0f46
jupyter_1      | or http://127.0.0.1:8888/?token=ac7f4a79a9dea63b36e08e179a9d7e61c084c8395f0d0f46
elasticsearch_1 | OpenJDK 64-Bit Server VM warning: Option UseConcMarkSweepGC was deprecated in version 9.0 and will like
ly be removed in a future release.
elasticsearch_1 | {"type": "server", "timestamp": "2025-06-16T09:12:17.536+0000", "level": "INFO", "component": "o.e.e.No
deEnvironment", "cluster.name": "docker-cluster", "node.name": "09ac8a4b9024", "message": "using [1] data paths, mounts [
/ (overlay)]", net usable_space [1.8gb], net total_space [15.4gb], types [overlay]}

```

```

ubuntu@ip-172-31-18-113:~/datascientest-data-engineering/docker/docker_compose_exercise$ docker ps
CONTAINER ID   IMAGE                                COMMAND                  CREATED        STATUS        PORTS
09ac8a4b9024   elasticsearch:7.2.0                "/usr/local/bin/dock-   6 minutes ago Up 6 minutes   9200/tcp,
9300/tcp
2013595c8b73   jupyter/minimal-notebook:ubuntu-18.04 "tini -g -- start-no-   6 minutes ago Up 6 minutes   0.0.0.0:44
44->8888/tcp, ::1444->8888/tcp
d0cc0bb420a28   postgres:16-alpine                "docker-entrypoint.s-   6 weeks ago   Up 37 minutes   0.0.0.0:54
32->5432/tcp, ::15432->5432/tcp
475951b12750   dpkg/pgadmin4                      "/entrypoint.sh"       6 weeks ago   Up 37 minutes   443/tcp, 0
.0.0.0:5050->80/tcp, ::15050->80/tcp
pgadmin4_container
ubuntu@ip-172-31-18-113:~/datascientest-data-engineering/docker/docker_compose_exercise$ docker-compose down
Stopping docker_compose_exercise_elasticsearch_1 ... done
Stopping docker_compose_exercise_jupyter_1 ... done
Removing docker_compose_exercise_elasticsearch_1 ... done
Removing docker_compose_exercise_jupyter_1 ... done
Removing network docker_compose_exercise_default
ubuntu@ip-172-31-18-113:~/datascientest-data-engineering/docker/docker_compose_exercise$

```

Edit docker-compose.yml file again (expose Elasticsearch ports):

```

docker-compose.yml M X
datascientest-data-engineering > docker > docker_compose_exercise
1  version: '3.9'
2  services:
3    jupyter:
4      image: jupyter/minimal-notebook:ubuntu-18.04
5      ports:
6        - target: 8888
7          published: 4444
8          protocol: tcp
9          mode: host
10   elasticsearch:
11     image: elasticsearch:7.2.0
12     environment:
13       - discovery.type: single-node
14     ports:
15       - "9200:9200"
16       - "9300:9300"

```

```

ubuntu@ip-172-31-18-113:~/datascientest-data-engineering/docker/docker_compose_exercise$ docker-compose up
Building with native build. Learn about native build in Compose here: https://docs.docker.com/go/compose-native-build/
Creating network "docker_compose_exercise_default" with the default driver
Creating docker_compose_exercise_jupyter_1 ... done
Creating docker_compose_exercise_elasticsearch_1 ... done
Attaching to docker_compose_exercise_jupyter_1, docker_compose_exercise_elasticsearch_1
jupyter_1      | Executing the command: jupyter notebook
jupyter_1      | [I 09:25:52.352 NotebookApp] Writing notebook server cookie secret to /home/jovyan/.local/share/jupyter
/runtime/notebook_cookie_secret
jupyter_1      | [I 09:25:53.616 NotebookApp] JupyterLab extension loaded from /opt/conda/lib/python3.7/site-packages/ju
pyterlab
jupyter_1      | [I 09:25:53.617 NotebookApp] JupyterLab application directory is /opt/conda/share/jupyter/lab
jupyter_1      | [I 09:25:53.623 NotebookApp] Serving notebooks from local directory: /home/jovyan
jupyter_1      | [I 09:25:53.624 NotebookApp] The Jupyter Notebook is running at:
jupyter_1      | http://b537a6678478:8888/?token=f71f280349c8000825ed35dc50778098a70401e2ca952088
jupyter_1      | [I 09:25:53.625 NotebookApp] http://b537a6678478:8888/?token=f71f280349c8000825ed35dc50778098a70401e2ca
952088
jupyter_1      | [I 09:25:53.626 NotebookApp] or http://127.0.0.1:8888/?token=f71f280349c8000825ed35dc50778098a70401e2c
a952088
jupyter_1      | [I 09:25:53.626 NotebookApp] Use Control-C to stop this server and shut down all kernels (twice to skip
confirmation).
jupyter_1      | [C 09:25:53.634 NotebookApp]
jupyter_1      |
jupyter_1      | To access the notebook, open this file in a browser:
jupyter_1      | file:///home/jovyan/.local/share/jupyter/runtime/nbserver-6--open.html
jupyter_1      | Or copy and paste one of these URLs:
jupyter_1      | http://b537a6678478:8888/?token=f71f280349c8000825ed35dc50778098a70401e2ca952088
jupyter_1      | or http://127.0.0.1:8888/?token=f71f280349c8000825ed35dc50778098a70401e2ca952088
elasticsearch_1 | OpenJDK 64-Bit Server VM warning: Option UseConcMarkSweepGC was deprecated in version 9.0 and will like
ly be removed in a future release.
elasticsearch_1 | {"type": "server", "timestamp": "2025-06-16T09:25:57.576+0000", "level": "INFO", "component": "o.e.e.No
deEnvironment", "cluster.name": "docker-cluster", "node.name": "fed9e8c3f822", "message": "using [1] data paths, mounts [
/ (overlay)]", net usable_space [1.8gb], net total_space [15.4gb], types [overlay]}

```

```

ubuntu@ip-172-31-18-113:~/datascientest-data-engineering/docker/docker_compose_exercise$ docker ps
CONTAINER ID   IMAGE                                COMMAND                  CREATED        STATUS        PORTS
fed9e0c3f822   elasticsearch:7.2.0                "/usr/local/bin/dock-   3 minutes ago Up 3 minutes   0.0.0.0:92
00->9200/tcp,  :::9200->9200/tcp, 0.0.0.0:9300->9300/tcp,  :::9300->9300/tcp   docker_compose_exercise_elasticsearch_1
b537a6678478   jupyter/minimal-notebook:ubuntu-18.04 "tini -g -- start-no-   3 minutes ago Up 3 minutes   0.0.0.0:44
44->8888/tcp,  :::4444->8888/tcp   docker_compose_exercise_jupyter_1
d6cc0b428d28   postgres:16-alpine                "docker-entrypoint.s-   6 weeks ago Up 47 minutes   0.0.0.0:54
32->5432/tcp,  :::5432->5432/tcp   pg_container
475951b12750   dpape/pgadmin4                     "/entrypoint.sh"        6 weeks ago Up 47 minutes   443/tcp, 0
.0.0.0:5050->80/tcp, :::5050->80/tcp   pgadmin4_container
ubuntu@ip-172-31-18-113:~/datascientest-data-engineering/docker/docker_compose_exercise$ docker-compose down
Stopping docker_compose_exercise_elasticsearch_1 ... done
Stopping docker_compose_exercise_jupyter_1 ... done
Removing docker_compose_exercise_elasticsearch_1 ... done
Removing docker_compose_exercise_jupyter_1 ... done
Removing network docker_compose_exercise_default
ubuntu@ip-172-31-18-113:~/datascientest-data-engineering/docker/docker_compose_exercise$

```

Edit docker-compose.yml file again (add custom container names + network):

```

docker-compose.yml M X
datascientest-data-engineering > docker > docker_compose_exe
1  version: '3.9'
   > Run All Services
2  services:
   > Run Service
3    jupyter:
4      image: jupyter/minimal-notebook:ubuntu-18.04
5      container_name: my_jupyter_from_compose
6      networks:
7        - my_network_from_compose
8      ports:
9        - target: 8888
10         published: 4444
11         protocol: tcp
12         mode: host
13      environment:
14        JUPYTER_TOKEN: 'hello'
   > Run Service
15    elasticsearch:
16      image: elasticsearch:7.2.0
17      container_name: my_es_from_compose
18      networks:
19        - my_network_from_compose
20      ports:
21        - "9200:9200"
22        - "9300:9300"
23      environment:
24        discovery.type: single-node
25      networks:
26        my_network_from_compose:

```

```

ubuntu@ip-172-31-18-113:~/datascientest-data-engineering/docker/docker_compose_exercise$ docker-compose up
Building with native build. Learn about native build in Compose here: https://docs.docker.com/go/compose-native-build/
Creating network "docker_compose_exercise_my_network_from_compose" with the default driver
Creating my_es_from_compose ... done
Creating my_jupyter_from_compose ... done
Attaching to my_jupyter_from_compose, my_es_from_compose
my_jupyter_from_compose | Executing the command: jupyter notebook
my_jupyter_from_compose | [I 09:35:17.327 NotebookApp] Writing notebook server cookie secret to /home/jovyan/.local/share/
jupyter/runtime/notebook_cookie_secret
my_jupyter_from_compose | [I 09:35:18.925 NotebookApp] JupyterLab extension loaded from /opt/conda/lib/python3.7/site-pack
ages/jupyterlab
my_jupyter_from_compose | [I 09:35:18.926 NotebookApp] JupyterLab application directory is /opt/conda/share/jupyter/lab
my_jupyter_from_compose | [I 09:35:18.932 NotebookApp] Serving notebooks from local directory: /home/jovyan
my_jupyter_from_compose | [I 09:35:18.933 NotebookApp] The Jupyter Notebook is running at:
my_jupyter_from_compose | [I 09:35:18.934 NotebookApp] http://1fcce2ccc1c8:8888/?token=...
my_jupyter_from_compose | [I 09:35:18.934 NotebookApp] or http://127.0.0.1:8888/?token=...
my_jupyter_from_compose | [I 09:35:18.935 NotebookApp] Use Control-C to stop this server and shut down all kernels (twice
to skip confirmation)
my_es_from_compose | OpenJDK 64-Bit Server VM warning: Option UseConcMarkSweepGC was deprecated in version 9.0 and will li
kely be removed in a future release.
my_es_from_compose | {"type": "server", "timestamp": "2025-06-16T09:35:22,753+0000", "level": "INFO", "component": "o.e.e.
NodeEnvironment", "cluster.name": "docker-cluster", "node.name": "2d3c1566f41a", "message": "using [1] data paths, mounts
[/ (overlay)]", net_usable_space [1.8gb], net_total_space [15.4gb], types [overlay]}

```

Open URL <http://54.78.195.186:4444/> and enter token, then create a notebook:

Jupyter interface showing the notebook content:

```

In [1]: import requests
import pprint
import json

content = json.loads(requests.get('http://my_es_from_compose:9200').content)
pprint.pprint(content)

{'cluster_name': 'docker-cluster',
 'cluster_uuid': 'y0--eFJ5q6Zw9KAgcWfuw',
 'name': '2d3c1566f41a',
 'tagline': 'You Know, for Search',
 'version': {'build_date': '2019-06-20T15:54:18.811730Z',
 'build_flavor': 'default',
 'build_hash': '508c3ba',
 'build_snapshot': False,
 'build_type': 'docker',
 'lucene_version': '8.0.0',
 'minimum_index_compatibility_version': '6.0.0-beta1',
 'minimum_wire_compatibility_version': '6.8.0',
 'number': '7.2.0'}}

```

```

ubuntu@ip-172-31-18-113:~/datascientest-data-engineering/docker/docker_compose_exercise$ docker ps
CONTAINER ID   IMAGE                                COMMAND                  CREATED        STATUS        PORTS
1f6ce2cc1c8    jupyter/minimal-notebook:ubuntu-18.04 "tini -g -- start-no..." 8 minutes ago  Up 8 minutes  0.0.0.0
4444->8888/tcp, 1::4444->8888/tcp
2d5c1566f415    elasticsearch:7.2.0                "/usr/local/bin/dock..." 8 minutes ago  Up 8 minutes  0.0.0.0
9200->9200/tcp, 1::9200->9200/tcp, 0.0.0.0:9300->9300/tcp, 1::9300->9300/tcp
my_es_from_compose
06c0b426d26    postgres:16-alpine                "docker-entrypoint.s..." 6 weeks ago   Up About an hour  0.0.0.0
5432->5432/tcp, 1::5432->5432/tcp
pg_container
475951b12750    dpape/pgadmin4                     "/entrypoint.sh"        6 weeks ago   Up About an hour  443/tcp
0.0.0.0:5888->88/tcp, 1::5888->88/tcp
pgadmin4_container
ubuntu@ip-172-31-18-113:~/datascientest-data-engineering/docker/docker_compose_exercise$ docker-compose down
Stopping my_jupyter_from_compose ... done
Stopping my_es_from_compose ... done
Removing my_jupyter_from_compose ... done
Removing my_es_from_compose ... done
Removing network docker_compose_exercise_my_network_from_compose
ubuntu@ip-172-31-18-113:~/datascientest-data-engineering/docker/docker_compose_exercise$

```

Edit `docker_compose.yml` file again (mount host volume into Jupyter):

```

docker-compose.yml M X
datascientest-data-engineering > docker > docker_compose_exercise >
1  version: '3.9'
2  services:
3    jupyter:
4      image: jupyter/minimal-notebook:ubuntu-18.04
5      container_name: my_jupyter_from_compose
6      networks:
7        - my_network_from_compose
8      volumes:
9        - ../home/jovyan/work
10     ports:
11       - target: 8888
12         published: 4444
13         protocol: tcp
14         mode: host
15     environment:
16       JUPYTER_TOKEN: 'bonjour'
17   elasticsearch:
18     image: elasticsearch:7.2.0
19     container_name: my_es_from_compose
20     networks:
21       - my_network_from_compose
22     ports:
23       - "9200:9200"
24       - "9300:9300"
25     environment:
26       discovery.type: single-node
27   networks:
28     my_network_from_compose:

```

```

ubuntu@ip-172-31-18-113:~/datascientest-data-engineering/docker/docker_compose_exercise$ docker-compose up
Building with native build. Learn about native build in Compose here: https://docs.docker.com/go/compose-native-build/
Creating network "docker_compose_exercise_my_network_from_compose" with the default driver
Creating my_jupyter_from_compose ... done
Creating my_es_from_compose ... done
Attaching to my_es_from_compose, my_jupyter_from_compose
my_jupyter_from_compose | Executing the command: jupyter notebook
my_jupyter_from_compose | I 09:49:45.794 NotebookApp] Writing notebook server cookie secret to /home/jovyan/.local/share/
my_jupyter_from_compose | JupyterLab extension loaded from /opt/conda/lib/python3.7/site-pack
my_jupyter_from_compose | I 09:49:47.822 NotebookApp] JupyterLab application directory is /opt/conda/share/jupyter/lab
my_jupyter_from_compose | I 09:49:47.829 NotebookApp] Serving notebooks from local directory: /home/jovyan
my_jupyter_from_compose | I 09:49:47.830 NotebookApp] The Jupyter Notebook is running at:
my_jupyter_from_compose | I 09:49:47.831 NotebookApp] http://0.0.0.0:4444/?token=...
my_jupyter_from_compose | I 09:49:47.832 NotebookApp] or http://127.0.0.1:8888/?token=...
my_jupyter_from_compose | I 09:49:47.832 NotebookApp] Use Control-C to stop this server and shut down all kernels (twice
my_jupyter_from_compose | ("type": "server", "timestamp": "2025-06-16T09:49:51.609+0000", "level": "INFO", "component": "o.e.e.
NodeEnvironment", "cluster_name": "docker-cluster", "node_name": "bfdf4ccca223", "message": "using [1] data paths, mounts
[/(overlay)], net usable_space [1.8gb], net total_space [15.4gb], types [overlay]" }
my_es_from_compose | ("type": "server", "timestamp": "2025-06-16T09:49:51.621+0000", "level": "INFO", "component": "o.e.e.
NodeEnvironment", "cluster_name": "docker-cluster", "node_name": "bfdf4ccca223", "message": "heap size [1007.3mb], compr
ssed ordinary object pointers [true]" }

```

Open URL <http://54.78.195.186:4444/> and enter token:



Navigate to <http://54.78.195.186:4444/tree/work> to check content of work folder:

